FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Main)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT – III
(Main)

	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	A
2.	B	B	D
3.	B	B	B
4.	D	D	B
5.	D	D	D
6.	A	A	B
7.	A	B	B
8.	B	C	C
9.	A	A	B
10.	A	B	C
11.	B	B	B
12.	C	B	D
13.	B	B	B
14.	C	A	B
15.	B	B	C
16.	B	B	C
17.	A	B	D
18.	B	B	C
19.	D	D	C
20.	A	B	A
21.	C	B	A
22.	A	D	C
23.	A	D	A
24.	D	C	A
25.	A	D	C
26.	C	A	D
27.	D	D	B
28.	C	C	B
29.	D	D	A
30.	C	B	A

Physics

PART – I

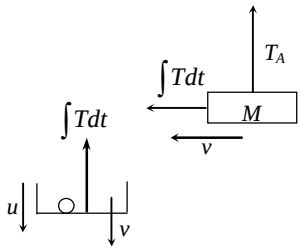
SECTION – A

1. $-\int T\,dt = Mv - Mu$ and $\int T\,dt = Mv$

$v = \frac{u}{2}$

$T_{A\,new} = Mg + M\frac{v^2}{l} = Mg + M\frac{u^2}{4l} = \frac{3Mg}{2}$

$T_{A\ old} = Mg$



2. Let equivalent resistance between *A* and *B* is *R*.

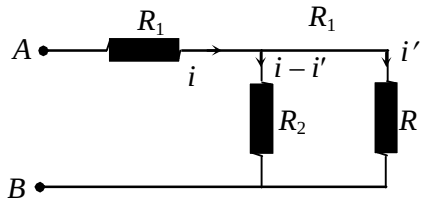
$iR = V, \quad i'R = \frac{V}{3} \quad \Rightarrow \quad i' = \frac{i}{3}$

$\frac{V}{3} = \frac{2i}{3}R_2$

$R = 2R_2$

$iR_1 = \frac{2V}{3}$

$\frac{R_1}{R_2} = \frac{4}{3}$



3. Conserving angular momentum

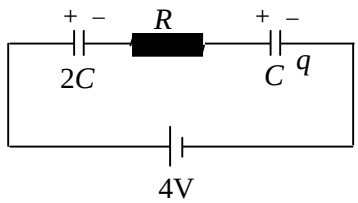
$mvR - \frac{3mR^2\omega}{2} = 3mR^2\omega'$

$\Rightarrow \quad \omega' = \frac{2v - 3R\omega}{6R}$

4. Let at any time a charge *q* flows through the circuit. The circuit is as shown

Time constant $(\tau) = \frac{2RC}{3}$

$\therefore \quad q = \frac{4CV}{3}\left(1 - e^{\frac{-3t}{2RC}}\right)$



5. $I_{M_1} = \frac{V_M}{\sqrt{R^2 + \left(\omega_1 L - \frac{1}{\omega_1 C}\right)^2}}$

Similarly, $I_{M_2} = \frac{V_M}{\sqrt{R^2 + \left(\omega_2 L - \frac{1}{\omega_2 C}\right)^2}}$

Since, both I_{M_1} and I_{M_2} are *n* times less than resonance current amplitude,

$\therefore \quad I_{M1} = I_{M2}$

$\therefore \quad \dfrac{V_M}{\sqrt{R^2 + \left(\omega_1 L - \dfrac{1}{\omega_1 C}\right)^2}} = \dfrac{V_M}{\sqrt{R^2 + \left(\omega_2 L - \dfrac{1}{\omega_2 C}\right)^2}}$

or $\omega_1 L - \dfrac{1}{\omega_1 C} = \dfrac{1}{\omega_2 C} - \omega_2 L$

or $\dfrac{\omega_1^2 LC - 1}{\omega_1 C} = \dfrac{1 - \omega_2^2 LC}{\omega_2 C}$

or $\omega_1\omega_2 LC(\omega_1 + \omega_2) - (\omega_1 + \omega_2) = 0$

or $1 - \omega_1\omega_2 LC = 0$

$\therefore \quad \omega_1\omega_2 = \dfrac{1}{LC} = \omega_0^2$

$\therefore \quad \omega_0 = \sqrt{\omega_1\omega_2}$

6. $f_0 = \dfrac{1}{2l}\sqrt{\dfrac{T}{\mu}}$

$\dfrac{df_0}{f_0} = \dfrac{1}{2}\dfrac{dT}{T}$

$f_0 = 2 \times 15\dfrac{T}{21} \times 100 = 150 \text{ Hz.}$

7. $W = \int PdV$ and $P \propto V$

9. The object distance for the mirror

$= -\left(15 + \dfrac{20}{4/3}\right) = -30 \text{ cm}$

For mirror,

$\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$

$\Rightarrow \quad \dfrac{1}{v} + \dfrac{1}{-30} = \dfrac{1}{-20} \Rightarrow v = -60 \text{ cm}$

The distance from water surface equals 60 – 15 = 45 cm

12. Least count = $\dfrac{\text{Pitch}}{\text{Number of divisions on circular scale}}$

$= \dfrac{1\text{mm}}{100} = 0.01\text{mm} = 0.001\text{cm}$

Zero error = – 0.03 mm

Measurement = 3.76 – (– 0.03) = 3.79 mm

13. $\vec{r}_{cm} = \dfrac{a}{2}\hat{i} + \dfrac{a}{2}\hat{j}$ (considering *A* to be origin *AD* to *x*-axis and *AB* to be *y*-axis)

$\vec{L}_i = m\vec{r}_{cm} \times \vec{v}_{cm} = m\left(\dfrac{a}{2}\hat{i} + \dfrac{a}{2}\hat{j}\right) \times v_0(2\hat{i} + 4\hat{j})$

$= mav_0(2\hat{k} - \hat{k}) = mav_0\hat{k}$

$$L_f = \left[\frac{m}{12}(a^2 + a^2) + m\frac{a^2}{2}\right]\omega = \left(\frac{ma^2}{6} + \frac{ma^2}{2}\right)\omega$$

From conservation of angular momentum about point *A*

$L_f = L_i$

$$\frac{2}{3}ma^2\omega = mav_0$$

$$\Rightarrow \quad \omega = \frac{3v_0}{2a}$$

$$|\vec{v}_{cm}| = \omega\frac{a}{\sqrt{2}} = \frac{3v_0}{2a}\frac{a}{\sqrt{2}} = \frac{3v_0}{2\sqrt{2}}$$

14. If V_0 be volume of the gas at *C*, $P_0V_0 = 3RT_0$

$$W = 3R(2T_0)\ln 2 - P_0V_0$$
$$= 6RT_0\ln 2 - 3RT_0$$
$$= 3RT_0[2\times 0.693 - 1]$$
$$= 3RT_0[1.386 - 1]$$
$$= 1.16RT_0$$

15. $mv_0R = I\omega + mvR$

$$= mR^2\left(\frac{v}{R}\right) + mvR$$

$$v = \frac{v_0}{2}$$

$$K_i = \frac{1}{2}mv_0^2$$

$$K_f = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}m\left(\frac{v_0}{2}\right)^2 + \frac{1}{2}mR^2\frac{v_0^2}{4R^2}$$

$$= \frac{1}{4}mv_0^2$$

So $W_f = -\frac{1}{4}mv_0^2$

16. Initial current in R_3, $i_i = \frac{E}{R_1 + R_3}$

Final current $i_f = \frac{E}{R_1 + \frac{R_2R_3}{R_2 + R_3}} \times \frac{R_2}{R_2 + R_3}$

$$= \frac{R_2E}{R_1(R_2 + R_3) + R_2R_3}$$

$$i_f = \frac{E}{\frac{R_1R_3}{R_2} + R_1 + R_3}$$

As $i_f < i_i$, so current in the resistor R_3 will decrease.

17. $\frac{dy}{dx} = -\frac{1}{\sqrt{3}} = \frac{v_y}{v_x}$

So $v_x = -v_y\sqrt{3}$

$v_x^2 + v_y^2 = v_0^2$

$3v_y^2 + v_y^2 = v_0^2$

$v_y = \pm\frac{v_0}{2}$

$v_x = \mp\frac{\sqrt{3}v_0}{2}$

So $y = \alpha e^{\frac{-2}{\sqrt{3}v_0}\times\frac{\sqrt{3}v_0}{2}}$

$y = \alpha e^{-1}$

18. If the string is pulled down by distance x, it is equivalent that a mass of λx is removed from the upper end of the string and is connected to the lower end

So $W_g = -\Delta U = U_i - U_f$

$W_g = \frac{-GM(\lambda x)}{R+L} - \left(\frac{-GM\lambda x}{R}\right)$

$= GM\lambda x\left[\frac{1}{R} - \frac{1}{R+L}\right]$

19. $a_{max} = \omega^2 A = \frac{K}{m}A$

If mass and amplitude both are doubled maximum acceleration will not change.

20. Resultant wave amplitude is given by

$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_1 - \varphi_2)}$

Value of $\cos(\phi_1 - \phi_2)$ lies between 1 and – 1

Hence $A_1 - A_2 \le A \le A, + A_2$

24. If three waves meet in same phase then the resultant amplitude will be algebraic addition of all three amplitudes

$R = 3A$

$A \rightarrow$ Amplitude of superimposed wave

$R \rightarrow$ Amplitude of resultant wave

than $I = 9I_0$ as $[I \propto R^2]$

25. If process starts and ends on same isotherm then product of P and V must be constant hence process a, b, d will be on same isotherm.

26. $\gamma_{mix} = \frac{n_1Cp_1 + n_2Cp_2}{n_1Cv_1 + n_2Cv_2}$

$= \frac{2\times\frac{7}{2}R + 1\times\frac{5}{2}R}{2\times\frac{5}{2}R + 1\frac{3}{2}R} = \frac{19}{13}$

27. $\frac{1}{\lambda_{K\alpha}} = R(Z-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$ .....(i)

$\frac{1}{\lambda'_{K\alpha}} = R(Z'-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$ ....(ii)

Solving (i) and (ii) $Z' = 9$

28. $\delta = A(\mu - 1)$ Slope of line will be A.

30. $\Delta x = \frac{\lambda}{2}$ $(SS_1 + S_1O) - (SO) = \frac{\lambda}{2}$

$2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}$

$2D\left[1+\frac{d^2}{2D^2}-1\right] = \frac{\lambda}{2}$

$d = \sqrt{\frac{\lambda D}{2}}$

Chemistry

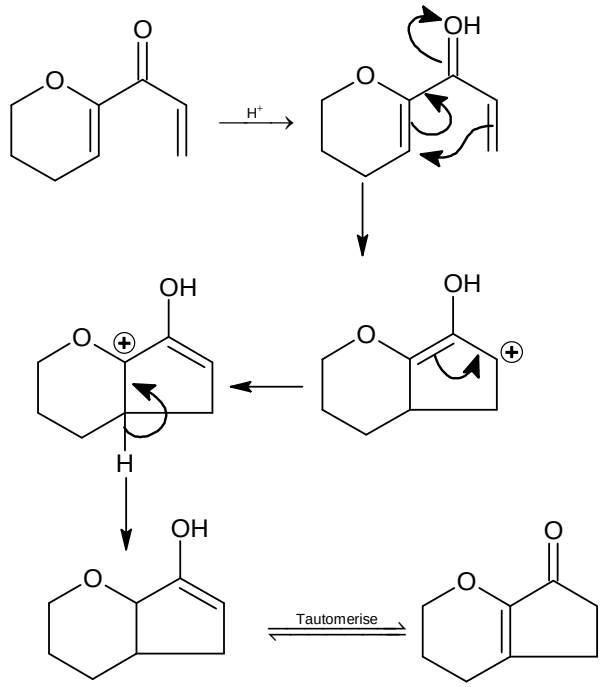
PART – II

SECTION – A

1. In gas phase inductive effect play a key role.

2. Geometrical isomers = $8 = 2^n$
   Optical isomer = $8 = 2^3 = 8$
   $= 16$

3. Due to bulky effect.

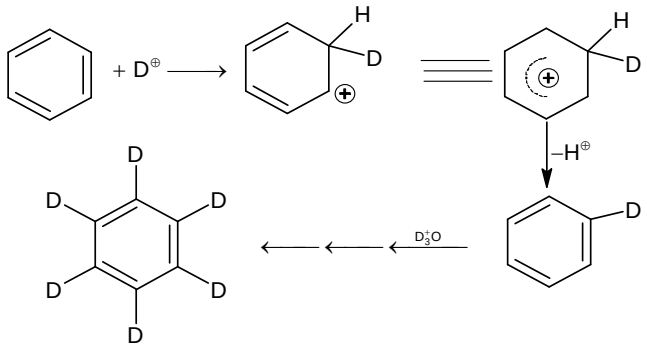
4. 

5. Birch reduction.

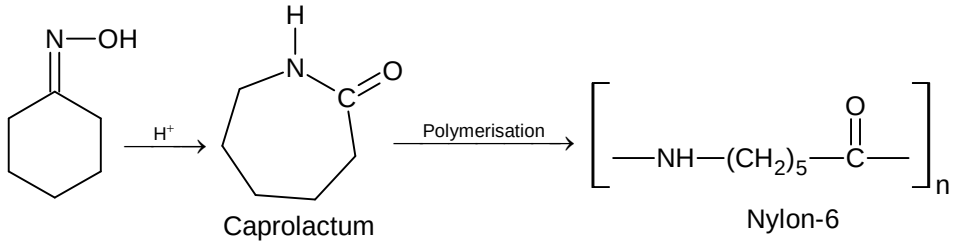
6. 3° carbocation is more stable.

7. $:CCl_2$ intermediate.

8. $CH_3CHO + 2CuO \longrightarrow CH_3COOH + \underset{\text{Red ppt.}}{Cu_2O} \downarrow$

9. 

10.

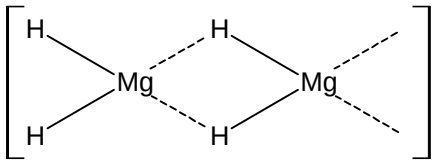


Caprolactum Nylon-6

11. It is belong to f-block – Lu element.

12. $IP_4 - IP_5$ required more energy, So IVA group elements.

15.



16. $3NaOH + 2O_3 \longrightarrow 2NaO_3 + NaOH.H_2O(s) + \frac{1}{2}O_2$

17. B.A $\underset{110}{Cl_2O},\ \underset{118}{ClO_2},\ \underset{105}{F_2O},\ \underset{112}{Br_2O}$

19. (II) and (IV) have unpaired electron.

20. $H_2S + CdSO_4 \longrightarrow \underset{\text{Yellow ppt}}{CdS}$

21. Let 1 mole of mixture has x mole N_2O_4
$2 \times 27.6 = x(92) + (1 - x)\ 46$
$x = 0.2$

22.

	2p	3p	3d
m =	+1	+1	+1
s =	+1/2	+1/2	+1/2

23. $M = \frac{\%\text{ by weight} \times 10 \times d}{M.w} = \frac{36.5 \times 10 \times 1.2}{36.5} = 12\ M$

$m = \frac{36.5 \times 1000}{36.5 \times (100 - 36.5)} = \frac{1000}{63.5} = 15.7\ m$

24. $ClO_3^- \longrightarrow ClO_4^- + 2e^-\ (x = 2)$ oxidation
$x = 5, \quad x = 7$
$e^- + ClO_3^- \longrightarrow ClO_2\ (x = 1)$ reduction
$x = 5,\ x = 4$
$E_W = M + \frac{M}{2}$

25. Ideal gas has no force of attraction and has negligible volume. Hence it can't be liquefied at any T and P.

26. $\Delta H = nxC_p \times \Delta T$ $C_p = C_v + K$

$n = \frac{PV}{RT} = 0.05$ $C_p = 20.794\ JK^{-1}$

For reversible adiable $(TV)^{r-1} = constant$

$T_2V_2^{r-1} = T_1V_1^{r-1}$ r = 1.66 for argon

$T_2 = 189.85$ k $\Delta T = -110.15$

Thus $\Delta H = 0.05 \times 20.794 \times -110.15) = -114.52$ J

27. $CH_3COCH_3 \rightleftharpoons CH_3CH_3(g) + CO(g)$

At $t = 0$	150	0	0
At $t = t$	$150 - x$	x	x

$\frac{x}{150 + x} = \frac{1}{3}$

x = 75 mm.

28. $Mg(OH)_2 \rightleftharpoons \underset{s}{Mg^{2+}} + \underset{2s}{2OH^-}$

$4s^3 = 8.9 \times 10^{-12}$,

$s = 1.3 \times 10^{-4}$ m

$\therefore$ $[OH] = 2 \times 1.305 \times 10^{-4}$

pOH = 3.58832

pH = 10.4168

29. 1 mole of A will form 2 moles of B and C after completion of reaction when 75%. A converted into B and C then total no. of moles

$0.5 + (2 \times 1.5) = 3.5$ moles

30. Along one body diagonal 2X atoms from 2 corners, one Y particle (at the centre of cube) will be removed.

So effective no. of X particle in a unit cell = $4 - 2 \times \frac{1}{8} = \frac{15}{4}$

And effective no. of Y particle in a unit cell = 4 – 1 = 3

$X : Y = \frac{15}{4} : 3$

5 : 4

Mathematics

PART – III

SECTION – A

1. $\alpha^n + \beta^n = -p^n$ ..... (1)

Again since $\frac{\alpha}{\beta}$ is a root of equation

$x^n + 1 + (x + 1)^n = 0$

$\left(\frac{\alpha}{\beta}\right)^n + 1 + \left(\frac{\alpha}{\beta} + 1\right)^n = 0$

$\alpha^n + \beta^n + (\alpha + \beta)^n = 0$

$\alpha^n + \beta^n = -(\alpha + \beta)^n = -(-p)^n$ ..... (2)

From equation (1) and (2), we get

$-p^n = -(-p)^n \Rightarrow p^n = (-p)^n$

$\Rightarrow p^n = (-1)^n p^n \Rightarrow (-1)^n = 1$

$\Rightarrow$ n is an even integer

2. From the figure, OM is bisector of angle $\angle$BOP of triangle OBP

$\therefore \frac{250}{50} = \frac{OB}{OP}$

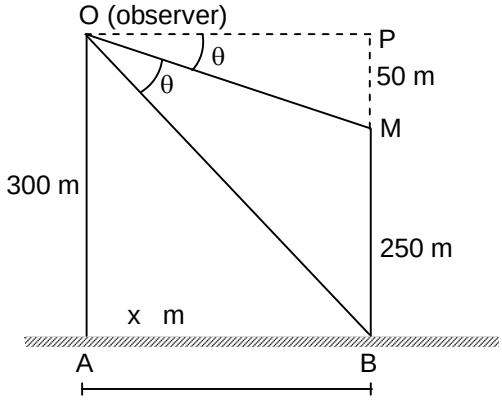
$\Rightarrow 5 = \frac{\sqrt{300^2 + x^2}}{x}$

$\Rightarrow 25x^2 - x^2 = (300)^2$

$\Rightarrow 24x^2 = (300)^2$

$\Rightarrow x^2 = \frac{300 \times 300}{24} = 3250 = 25 \times 25 \times 6$

$\Rightarrow x = 25\sqrt{6}.$



3. $P = 10^1 \cdot 10^{\frac{1}{2}} \cdot 10^{\frac{1}{2^2}} \cdot 10^{\frac{1}{2^3}} \cdot .....$

$P = 10^{1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+....\infty} = 10^{\frac{1}{1-\frac{1}{2}}} = 10^2 = 100$

$\log_{0.01} P = \log_{0.01}(100) = -1$

4. $\frac{ax^2 + b}{2} \le \sqrt{ax^2 b} = |x|\sqrt{ab}$

Consequently, for all x > 0

$y = \frac{x}{ax^2 + b} \le \frac{x}{2x\sqrt{ab}} = \frac{1}{2\sqrt{ab}}$

5. $\bar{x} = \frac{0.1 + 1 \cdot {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \cdots + n {}^nC_n}{1 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n} = \sum_{r=0}^{n} r {}^nC_r = \frac{\sum_{r=1}^{n} r \frac{n^{n-1}}{r} C_{r-1}}{\sum_{r=0}^{n} {}^nC_r}$

$= \frac{n\sum_{r=1}^{n} {}^{n-1}C_{r-1}}{\sum_{r=0}^{n} {}^nC_r} = \frac{n2^{n-1}}{2^n} = \frac{n}{2}$

6. Greatest coefficient is $\dfrac{n!}{(q!)^{m-r}\left((q+1)!\right)^{r}} = \dfrac{15!}{(3!)(4!)^{3}}$

$l^n = 15$ and $m = 4$, $q = 3$ and $r = 37$

$\because\ 15 = 4 \times 3 - 13$

7. Two triangles may have equal areas

$\therefore$ f is not one-one

Since each positive real number can represent area of a triangle

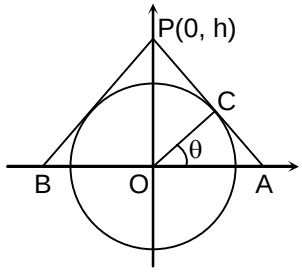
$\therefore$ f is onto

8. Area of $\Delta PAB = \dfrac{32}{\sin 2\theta}$

$(\Delta PAB)_{min}$ when $2\theta = \dfrac{\pi}{2}$

$\Rightarrow\ \theta = \dfrac{\pi}{4}$

$\Rightarrow$ h = OC cosec $\theta = 4\sqrt{2}$



9. Equation of chord joining α and β is $\dfrac{x}{a}\cos\left(\dfrac{\alpha-\beta}{2}\right) - \dfrac{y}{b}\sin\left(\dfrac{\alpha+\beta}{2}\right) = \cos\left(\dfrac{\alpha+\beta}{2}\right)$

$\therefore\ \alpha + \beta = 3\pi$

$\dfrac{x}{a}\cos\left(\dfrac{\alpha-\beta}{2}\right) + \dfrac{y}{b} = 0$

It passes through the centre (0, 0)

10. Ordinate of the point of intersection of the line $\dfrac{x}{a} - \dfrac{y}{b} = m$ and the hyperbola, given by

$\left(\dfrac{x}{a} - \dfrac{y}{b}\right)\left(\dfrac{x}{a} - \dfrac{y}{b} + \dfrac{2y}{b}\right) = 1$ i.e. $m\left(m + \dfrac{2y}{b}\right) = 1$

i.e. $y = \dfrac{b\left(1-m^2\right)}{2m}$

Similarly ordinate of the point of intersection of the line $\dfrac{x}{a} + \dfrac{y}{b} = m$ and the hyperbola is given by

$y = \dfrac{b\left(m^2-1\right)}{2m}$

Sum of the ordinates is zero

11. Let $P(x_1, y_1)$, $Q(x_2, y_2)$ be the given points on $y = 2^{x+2}$

$\therefore\ y_1 = 2^{x_1+2}$ and $y_2 = 2^{x_2+2}$

$\overrightarrow{OP}\cdot\hat{i}$ = projection of $\overrightarrow{OP}$ on x-axis = $x_1 = -1 \Rightarrow y_1 = 2$

and $\overrightarrow{OQ}\cdot\hat{i}$ = projection of $\overrightarrow{OQ}$ on x-axis = $x_2 = 2 \Rightarrow y_2 = 16$

$\therefore\ \overrightarrow{OP} = -\hat{i} + 2\hat{j}$, $\overrightarrow{OQ} = 2\hat{i} + 16\hat{j}$

$\therefore\ \overrightarrow{OQ} - 4\overrightarrow{OP} = 6\hat{i} + 8\hat{i}$

$\therefore\ \left|\overrightarrow{OQ} - 4\overrightarrow{OP}\right| = \left|6\hat{i} + 8\hat{j}\right| = 10$

12. The point B is (–q, –p) and the point C is (–p, –q) mid-point of AC is (0, 0)

13. The point Q is (2, 1)

Image of P(1, 2) in the line x – 2y + 1 = 0 is given by $\frac{x-1}{1}=\frac{y-2}{-2}=\frac{4}{5}$

$\therefore$ Ordinate of the point are $\left(\frac{9}{5},\frac{2}{5}\right)$

Since this point lies on QR

$\therefore$ Equation of QR is 3x – y – 5 = 0

$\therefore \lambda + \mu = 2$

14. Let $(\alpha, 3-\alpha)$ be any point on x + y = 3

$\therefore$ Equation of chord of contact is $\alpha x + (3-\alpha)y = 9$

i.e. $\alpha(x-y) + 3y - 9 = 0$

The chord passes through the point (3, 3) for all values of α

15. Equation of the two circles be $(x-r)^2 + (y-r)^2 = r^2$ i.e. $x^2 + y^2 - 2rx - 2ry + r^2 = 0$ where $r = r_1$ and r_2 condition of orthogonally gives $2r_1r_2 + 2r_1r_2 = r_1^2 + r_2^2$

$\Rightarrow 4r_1r_2 = r_1^2 + r_2^2$, circle passes through (a, b)

$\Rightarrow a^2 + b^2 - 2ra - 2rb + r^2 = 0$

i.e. $r^2 - 2r(a+b) + a^2 + b^2 = 0$

$\therefore 4(a^2+b^2) = 4(a+b)^2 - 2(a^2+b^2)$

$\Rightarrow a^2 - 4ab + b^2 = 0$

16. $y^2 = 6\left(x-\frac{3}{2}\right)$

Equation of directrix x = 0

Let coordinate P be $\left(\frac{3}{2}+\frac{3}{2}t^2, 3t\right)$

$\therefore$ Coordinate of M are (0, 3t)

$\therefore$ $MS = \sqrt{9+9t^2}$

$MP = \frac{3}{2}+\frac{3}{2}t^2$

$\therefore 9 + 9t^2 = \left(\frac{3}{2}+\frac{3}{2}t^2\right)^2 = \frac{9}{4}\left(1+t^2\right)^2$

$\therefore 4 = 1 + t^2$

$\Rightarrow$ Length of side = 6

17. $yy_1 = 2a(x + x_1)$

$x^2 = 4by = 4b\left[\left(\frac{2a}{y_1}\right)(x+x_1)\right]$

$\Rightarrow y_1x^2 - 8ab\,x - 8ab\,x_1 = 0$

D = 0 gives xy = –2ab

18. Equation of the tangent to the ellipse at P(5 cos θ, 4 sin θ) is $\frac{x\cos\theta}{5}+\frac{y\sin\theta}{4}=1$

It meets the line x = 0 at Q(0, 4 cosec θ)
Image of in the line y = x is R(4 cosec θ, 0)
Equation of the circle is $x(x-4\operatorname{cosec}\theta)+y(y-4\operatorname{cosec}\theta)=0$
i.e. $x^2+y^2-4(x+y)\operatorname{cosec}\theta=0$
$\therefore$ Each member of the family passes through the intersection of $x^2+y^2=0$ and $x+y=0$
i.e. The point (0, 0)

19. Equation of normal at (13 cos θ, 5 sin θ) of $\frac{x^2}{169}+\frac{y^2}{25}=1$

$(y-5\sin\theta)=\frac{13\sin\theta}{5\cos\theta}(x-13\cos\theta)$ it passes through (0, 6) then cos θ(15 + 72 sin θ) = 0

$\Rightarrow \cos\theta = 0,\ \sin\theta=-\frac{5}{24}$

$\Rightarrow \pi+\sin^{-1}\left(\frac{5}{24}\right)$

21. Given, f(x) = (x – 1)(x – 2)(x – 3)(x – 4)
Consider the interval [1, 2]
Since, f(x) is a polynomial function, it is continuous in [1, 2] and differentiable in (1, 2)
Also, f(1) = 0 = f(2)
By Rolle's theorem these exists a point x = c in (1, 2) such that f′(c) = 0 i.e. these exists a root of f′(x) = 0 in the interval (1, 2) which is positive
Similarly, two positive roots of f′(x) = 0 exist in the open interval (2, 3) and (3, 4)
Hence, the three roots of f′(x) are positive

22. f(x) g(y) + f′(x) g(y) = g′(y)
$\Rightarrow$ (f(x) + f′(x)) g(y) = g′(y)
$\Rightarrow f(x)+f'(x)=\frac{g'(y)}{g(y)}$ = constant ..... (1)

Put y = 0 $\Rightarrow \frac{g'(0)}{g(0)}=\frac{1}{1}$

Now from equation (1), we get f(x) + f′(x) = 1 and $\frac{g'(y)}{g(y)}=1$

$\Rightarrow g(y)=e^y$

23. ${}^{10}C_1+{}^{10}C_2+{}^{10}C_3+{}^{10}C_4=10+45+120+210=385$

24. Total number of numbers without restriction two numbers have all the digits equal
Hence, the required number of numbers = $2^5 - 2 = 30$

25. Given 7a – 9b = 0 $\Rightarrow b=\frac{7}{9}a$

So, the number of pairs (a, b) can be (9, 7), (18, 14) (27, 21) and (36, 28)

Required probability = $\frac{4}{{}^{39}C_2}=\frac{4}{741}$

26. Two numbers are selected from {1, 2, 3, 4, 5, 6}
$\therefore$ n(S) = 6×5 (as one by one with out replacement) favourable events = minimum of the two numbers is less than 4
$\therefore$ n(E) = 6×4
As for the minimum of the two is less then 4, we can select one from {1, 2, 3, 4} and other from {1, 2, 3, 4, 5, 6}
$\therefore$ Required probability = $\frac{n(E)}{n(S)} = \frac{24}{30} = \frac{4}{5}$

27. $\int \frac{dx}{\sqrt{\sin^4 x \cot x}} = \int \frac{dx}{\sin^2 x \sqrt{\cot x}} = \int \frac{\operatorname{cosec}^2 x}{\sqrt{\cot x}} dx$
Put $\cot x = t^2$
$\operatorname{cosec}^2 x\, dx = -2t\, dt$
$= -\int \frac{2t}{t} dt = -2t + c = -2\sqrt{\cot x} + c = -\frac{2}{\sqrt{\tan x}} + c$

28. Let $a = 2^x$, $b = -3^{x-1}$, $c = -1$
$\Rightarrow a^3 + b^3 + c^3 = 3abc$
Now, $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\Sigma(a-b)^2$ factorizing and get $a + b + c = 0$ as $a \neq b \neq c$
Hence, the given equation is; $2^x - 3^{x-1} - 1 = 0$
By trial x = 1 and 2 are solution of this
Now, $2^x = 3^{x-1} + 1$ has no other solution
Number of real solution exactly two

29. $2\sin^2\theta - 5\sin\theta + 2 > 0$
$\sin\theta < \frac{1}{2}$ ($\because -1 \le \sin\theta \le 1$)
$\theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

30. Let $h(x) = f^2(x) + (f'(x))^2$
$\Rightarrow h'(x) = -2x(f'(x))^2 g(x)$
$\Rightarrow x = 0$ is the point of maximum